

Lois Orosa Nogueira

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Interests

Computer Architecture

Education

University of Santiago de Compostela
in Computer Architecture

*Santiago de Compostela,
Spain*

Dec 2008 – Sept 2013

- Thesis: New Hardware Support for Transactional Memory and Parallel Debugging in Multicore Processors
- Advisor: Elisardo Antelo and Javier Bruguera
- Supported by the project "Hardware and software support for high performance computing" (TIN201017541)
- Interuniversity PhD Courses in Information Technology (University of Santiago de Compostela and University of A Coruña)

University of Vigo
M.Eng in Telecommunications Engineering

Vigo, Spain

Sept 2000 – June 2006

Experience

Scientific Director
Galicia Supercomputing Center (CESGA)

*Santiago de Compostela,
Spain*

Sept 2025 – present

- Scientific Director
- Designing a Scientific Strategy for scaling the institution
- Institutional representative
- Get and lead the European AI Factory 1HealthAI (82M EUR)

Senior Researcher
Galicia Supercomputing Center (CESGA)

*Santiago de Compostela,
Spain*

Sept 2025 – present

Design Reliable, secure and efficient computing systems for AI and other emerging applications

- Ramón y Cajal tenure track contract (ranked 1st with a score of 100/100)
- Leading a Consolidación Investigadora project (199.927 EUR.)
- Xunta the Galicia Consolidator Grant (115.000 EUR.)
- Leading the Experimental platform of the 1HealthAI AI Factory (4M EUR.)

Managing Director
Galicia Supercomputing Center (CESGA)

*Santiago de Compostela,
Spain*

Mar 2022 – Sept 2025

- Leading CESGA's strategic change.– Contributing to attract large European Funds for infrastructure, services and research.
- CESGA Representative in the Galicia Quantum Technology Hub (2022-2030, 154M EUR)
- Quantum Computer with more Qubits in South Europe (2023, 32 Qubits, 14M EUR)
- Second largest Supercomputer in Spain (Finisterrae III, 2022, 7M EUR)

Senior Researcher

ETH Zürich, D-ITET, SAFARI group

Memory Systems and Hardware Security

- PI: Onur Mutlu

Zurich, Switzerland

Jan 2019 – Mar 2022

Internship

Xilinx Research

Training Convolutional Neural Networks (CNNs)

- Mentors: Michaela Blott, Yaman Umuroglu

Dublin, Ireland

July 2018 – Dec 2018

PostDoctoral Fellow

University of Campinas (UNICAMP), LSC

Architectural Support for Speculative Program Execution

- Supported by FAPESP grant 2014/03840-2.
- PI: Rodolfo Azevedo.

Campinas, Brazil

June 2014 – July 2018

Academic Guest

ETH Zürich, Department of Computer Science, SAFARI Group

Enabling security features in commodity DRAM chips

- Supported by FAPESP grant 2016/18929-4.
- Supervisor: Onur Mutlu

Zürich, Switzerland

Jan 2017 – Dec 2017

PostDoctoral Researcher

University of Santiago de Compostela

- Supervisor: Elisardo Antelo and Javier Bruguera

Santiago de Compostela,

Spain

Jan 2014 – May 2014

PostDoctoral Researcher

Universidade Nova de Lisboa

Hardware support for detecting atomicity violations.

- Supported by European Cooperation in Science and Technology (COST) with a Short-term Scientific Mission (STSM).
- Supervisor: João Lourenço

Lisboa, Portugal

Sept 2013 – Dec 2013

Internship

Recore Systems

Development of a shared memory multicore simulator.

- Mentor: Gerard Rauwerda

Dublin, Ireland

Aug 2012 – Dec 2012

Summer Internship

IBM R&D Labs

Development of compiler-based deterministic replay for X10 Language

- Mentors: Olga Golovanevsky, Marina Biberstein, Bilha Mendelson

Dublin, Ireland

July 2010 – Nov 2010

Scholarship

University of Illinois at Urbana-Champaign (UIUC), U.S.A.

Tolerating concurrency bugs in multicore processors

- Department of Computer Science, IACOMA group
- Advisor: Josep Torrellas

Urbana-Champaign, U.S.A

Sept 2009 – Dec 2009

R&D Engineer

Arantia 2010 (Televes Group)

Development of network multimedia applications (TV streaming, multimedia contents).

Santiago de Compostela,

Spain

2006

Engineer

Communitel Global S.A (Vodafone)

Resolve second level maintenance issues, and automatize large-scale production tasks with perl and C.

Vigo, Spain

Apr 2006 – Oct 2006

Engineer

Communitel Global S.A (Vodafone)

Development of a web-based application for monitoring and controlling the network infrastructure.

Vigo, Spain

July 2005 – Oct 2005

Publications

SpyHammer: Understanding and exploiting RowHammer under fine-grained temperature variations

ACCESS'24

Orosa, Lois, Ruhrmair, Ulrich, Yaglikci, A Giray, Luo, Haocong, Olgun, Ataberk, Jattke, Patrick, Patel, Minesh, Kim, Jeremie S, Razavi, Kaveh, Mutlu, Onur
(IEEE access)

DRAM bender: An extensible and Versatile FPGA-based Infrastructure to Easily Test State-of-the-art DRAM chips

TCAD'23

Olgun, Ataberk, Hassan, Hasan, Yaglikci, A Giray, Tugrul, Yahya Can, Orosa, Lois, Luo, Haocong, Patel, Minesh, Ergin, Oguz, Mutlu, Onur
(IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD'23))

Using Approximate DRAM for Enabling Energy-Efficient, High-Performance Deep Neural Network Inference

Springer'23

Orosa, Lois, Koppula, Skanda, Kanellopoulos, Konstantinos, Yaglikci, A Giray, Mutlu, Onur
(Embedded Machine Learning for Cyber-Physical, IoT, and Edge Computing: Hardware Architectures, chapter 10)

Alp: Alleviating cpu-memory data movement overheads in memory-centric systems

TETC'22

Ghiasi, Nika Mansouri, Vijaykumar, Nandita, Oliveira, Geraldo F, Orosa, Lois, Fernandez, Ivan, Sadrosadati, Mohammad, Kanellopoulos, Konstantinos, Hajinazar, Nastaran, Luna, Juan Gomez, Mutlu, Onur
(IEEE Transactions on Emerging Topics in Computing (TETC'22))

EcoFlow: Efficient Convolutional Dataflows on Low-Power Neural Network Accelerators

TC'23

Orosa, Lois, Koppula, Skanda, Kanellopoulos, Konstantinos, Umuroglu, Yaman, Kanellopoulos, Konstantinos, Gomez-Luna, Juan, Blott, Michaela, Vissers, Kees, Mutlu, Onur
(IEEE Transactions on Computers (TC'23))

Understanding Rowhammer under Reduced Wordline Voltage: An experimental study using real dram devices

DSN'22

Yaglikci, A Giray, Luo, Haocong, De Oliveira, Geraldo F, Olgun, Ataberk, Patel, Minesh, Park, Jisung, Hassan, Hasan, Kim, Jeremie S, Orosa, Lois, Mutlu, Onur
(52nd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN'22))

pLUTo: Enabling Massively Parallel Computation in DRAM via Lookup Tables

MICRO'22

Ferreira, Joao Dinis, Falcao, Gabriel, Gomez-Luna, Juan, Alser, Mohammed, Orosa, Lois, Sadrosadati, Mohammad, Kim, Jeremie S, Oliveira, Geraldo F, Shahroodi, Taha, Nori, Anant, Mutlu, Onur
(55th IEEE/ACM International Symposium on Microarchitecture (MICRO'22))

Hira: Hidden Row Activation for Reducing Refresh Latency of Off-The-DRAM Chips

MICRO'22

Yaglikci, A Giray, Olgun, Ataberk, Patel, Minesh, Luo, Haocong, Hassan, Hasan, Orosa, Lois, Mutlu, Onur
(55th IEEE/ACM International Symposium on Microarchitecture (MICRO'22))

- NEON: Enabling Efficient Support for Nonlinear Operations in Resistive RAM-based Neural Network Accelerators** arXiv'22
Manglik, Aditya, Patel, Minesh, Mao, Haiyu, Salami, Behzad, Park, Jisung, Orosa, Lois, Mutlu, Onur
(arXiv preprint arXiv:2211.05730)
- Optically Connected Memory for Disaggregated Data Centers** JPDC'22
Gonzalez, Jorge, Palma, Mauricio G, Hattink, Maarten, Rubio-Noriega, Ruth, Orosa, Lois, Mutlu, Onur, Bergman, Keren, Azevedo, Rodolfo
(Journal of Parallel and Distributed Computing (JPDC'22))
- DR-STRaNGe: End-to-end System Design for DRAM-based True Random Number Generators** HPCA'22
Bostanci, F Nisa, Olgun, Ataberk, Orosa, Lois, Yaglikci, A Giray, Kim, Jeremie S, Hassan, Hasan, Ergin, Oguz, Mutlu, Onur
(28th IEEE International Symposium on High-Performance Computer Architecture (HPCA'22))
- IChannels: Exploiting Current Management Mechanisms to Create Covert Channels in Modern Processors** ISCA'21
Haj-Yahya, Jawad, Orosa, Lois, Kim, Jeremie S, Gomez-Luna, Juan, Yaglikci, A Giray, Alser, Mohammed, Puddu, Ivan, Mutlu, Onur
(48th IEEE International Symposium on Computer Architecture (ISCA'21))
- A Deeper Look into RowHammer's sensitivities: Experimental analysis of real DRAM Chips and Implications on Future Attacks and Defenses** MICRO'21
Orosa, Lois, Yaglikci, Abdullah Giray, Luo, Haocong, Olgun, Ataberk, Park, Jisung, Hassan, Hasan, Patel, Minesh, Kim, Jeremie S, Mutlu, Onur
(54th IEEE/ACM International Symposium on Microarchitecture (MICRO'21))
- Reducing Solid-state Drive Read Latency by Optimizing Read-retry** ASPLOS'21
Park, Jisung, Kim, Myungsuk, Chun, Myoungjun, Orosa, Lois, Kim, Jihong, Mutlu, Onur
(26th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'21))
- SynCron: Efficient Synchronization Support for Near-data-processing Architectures** HPCA'21
Giannoula, Christina, Vijaykumar, Nandita, Papadopoulou, Nikela, Karakostas, Vasileios, Fernandez, Ivan, Gomez-Luna, Juan, Orosa, Lois, Koziris, Nectarios, Goumas, Georgios, Mutlu, Onur
(27th IEEE International Symposium on High-Performance Computer Architecture (HPCA'21))
- Blockhammer: Preventing rowhammer at low cost by blacklisting rapidly-accessed dram rows** HPCA'21
Yaglikci, A Giray, Patel, Minesh, Kim, Jeremie S, Azizi, Roknoddin, Olgun, Ataberk, Orosa, Lois, Hassan, Hasan, Park, Jisung, Kanellopoulos, Konstantinos, Shahroodi, Taha, Ghose, Saugata, Mutlu, Onur
(27th IEEE International Symposium on High-Performance Computer Architecture (HPCA'21))
- DAMOV: A new methodology and benchmark suite for evaluating data movement bottlenecks** ACCESS'21
Oliveira, Geraldo F, Gomez-Luna, Juan, Orosa, Lois, Ghose, Saugata, Vijaykumar, Nandita, Fernandez, Ivan, Sadrosadati, Mohammad, Mutlu, Onur
(IEEE Access)

- Codic: A low-cost substrate for enabling custom in-dram functionalities and optimizations** ISCA'21
 Orosa, Lois, Wang, Yaohua, Sadrosadati, Mohammad, Kim, Jeremie S, Patel, Minesh, Puddu, Ivan, Luo, Haocong, Razavi, Kaveh, Gomez-Luna, Juan, Hassan, Hasan, others
 (48th IEEE International Symposium on Computer Architecture (ISCA'21))
- FIGARO: Improving System Performance via Fine-grained in-DRAM Data Relocation and Caching** MICRO'20
 Wang, Yaohua, Orosa, Lois, Peng, Xiangjun, Guo, Yang, Ghose, Saugata, Patel, Minesh, Kim, Jeremie S, Gomez-Luna, Juan, Sadrosadati, Mohammad, Ghiasi, Nika Mansouri, Mutlu, Onur
 (53th IEEE/ACM International Symposium on Microarchitecture (MICRO'20))
- Revisiting rowhammer: An Experimental Analysis of Modern DRAM Devices and Mitigation Techniques** ISCA'20
 Kim, Jeremie S, Patel, Minesh, Ya{\u{g}}l{\i}k{\c{c}}{\i}, A Giray, Hassan, Hasan, Azizi, Roknoddin, Orosa, Lois, Mutlu, Onur
 (47th Annual International Symposium on Computer Architecture (ISCA'20))
- FlexWatts: A Power-and Workload-aware Hybrid Power Delivery Network for Energy-efficient Microprocessors** MICRO'20
 Haj-Yahya, Jawad, Alser, Mohammed, Kim, Jeremie S, Orosa, Lois, Rotem, Efraim, Mendelson, Avi, Chattopadhyay, Anupam, Mutlu, Onur
 (53rd IEEE/ACM International Symposium on Microarchitecture (MICRO'20))
- WoLFraM: Enhancing Wear-leveling and Fault Tolerance in Resistive Memories using Programmable Address Decoders** ICCD'20
 Yavits, Leonid, Orosa, Lois, Mahar, Suyash, Ferreira, Joao Dinis, Erez, Mattan, Ginosar, Ran, Mutlu, Onur
 (38th IEEE International Conference on Computer Design (ICCD'20))
- Evanesco: Architectural Support for Efficient Data Sanitization in Modern Flash-based Storage Systems** ASPLOS'20
 Kim, Myungsuk, Park, Jisung, Cho, Genhee, Kim, Yoona, Orosa, Lois, Mutlu, Onur, Kim, Jihong
 (25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'20))
- Robust Machine Learning Systems: Challenges, Current Trends, Perspectives, and the Road Ahead** D&T'20
 Shafique, Muhammad, Naseer, Mahum, Theocharides, Theocharis, Kyrkou, Christos, Mutlu, Onur, Orosa, Lois, Choi, Jungwook
 (IEEE Design & Test (D&T))
- CLR-DRAM: A Low-cost DRAM Architecture Enabling Dynamic Capacity-latency Trade-off** ISCA'20
 Luo, Haocong, Shahroodi, Taha, Hassan, Hasan, Patel, Minesh, Yaglikci, A Giray, Orosa, Lois, Park, Jisung, Mutlu, Onur
 (47th Annual International Symposium on Computer Architecture (ISCA'20))
- D-RaNGe: Using commodity DRAM Devices to Generate True Random Numbers with Low Latency and High Throughput** HPCA'19
 Kim, Jeremie S, Patel, Minesh, Hassan, Hasan, Orosa, Lois, Mutlu, Onur
 (5th IEEE International Symposium on High-Performance Computer Architecture (HPCA'19))

- ITAP: Idle-time-aware Power Management for GPU Execution Units** TACO'19
Sadrosadati, Mohammad, Ehsani, Seyed Borna, Falahati, Hajar, Ausavarungnirun, Rachata, Tavakkol, Arash, Abaee, Mojtaba, Orosa, Lois, Wang, Yaohua, Sarbazi-Azad, Hamid, Mutlu, Onur
(ACM Transactions on Architecture and Code Optimization (TACO))
- EDEN: Enabling Energy-efficient, High-performance Deep Neural Network Inference using Approximate DRAM** MICRO'19
Koppula, Skanda, Orosa, Lois, Yaglikci, A Giray, Azizi, Roknoddin, Shahroodi, Taha, Kanellopoulos, Konstantinos, Mutlu, Onur
(52nd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO'19))
- Reducing DRAM Latency via Charge-level-aware Look-ahead partial Restoration** MICRO'18
Wang, Yaohua, Tavakkol, Arash, Orosa, Lois, Ghose, Saugata, Ghiasi, Nika Mansouri, Patel, Minesh, Kim, Jeremie S, Hassan, Hasan, Sadrosadati, Mohammad, Mutlu, Onur
(51st Annual IEEE/ACM International Symposium on Microarchitecture (MICRO'18))
- AVPP: Address-first Value-next Predictor with Value Prefetching for Improving the Efficiency of Load Value Prediction** TACO'18
Orosa, Lois, Azevedo, Rodolfo, Mutlu, Onur
(ACM Transactions on Architecture and Code Optimization (TACO))
- FLIN: Enabling Fairness and Enhancing Performance in Modern NVMe Solid State Drives** ISCA'18
Tavakkol, Arash, Sadrosadati, Mohammad, Ghose, Saugata, Kim, Jeremie, Luo, Yixin, Wang, Yaohua, Ghiasi, Nika Mansouri, Orosa, Lois, Gomez-Luna, Juan, Mutlu, Onur
(45th International Symposium on Computer Architecture (ISCA'18).)
- Asymmetric Allocation in a Shared Flexible Signature Module for Multicore Processors** CJ'16
Orosa, Lois, Bruguera, Javier D, Antelo, Elisardo
(The Computer Journal (CJ'16))
- A Hardware Approach to Detect, Expose and Tolerate High Level Data Races** PDP'16
Orosa, Lois, Lourenco, Joao
(4th Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP'16))
- Temporal frequent value locality** ICECS'16
Orosa, Lois, Azevedo, Rodolfo
(3rd IEEE International Conference on Electronics, Circuits and Systems (ICECS'16))
- Architecting a Computer with a Full Optical RAM** ICECS'16
Gonzalez, Jorge, Orosa, Lois, Azevedo, Rodolfo
(3rd IEEE International Conference on Electronics, Circuits and Systems (ICECS'16))
- Flexsig: Implementing Flexible Hardware Signatures** TACO'12
Orosa, Lois, Antelo, Elisardo, Bruguera, Javier D
(ACM Transactions on Architecture and Code Optimization (TACO))
- Pacman: Tolerating Asymmetric Data Races with Unintrusive Hardware** HPCA'12
Qi, Shanxiang, Otsuki, Norimasa, Nogueira, Lois Orosa, Muzahid, Abdullah, Torrellas, Josep
(8th International Symposium on High Performance Computer Architecture (HPCA'12))

Workshops

- LogSI-HTM: Log Based Snapshot Isolation in Hardware Transactional Memory** WTTM'15
Orosa, Lois, Azevedo, Rodolfo
(7th Workshop on the Theory of Transactional Memory (WTTM'15))
- A Hardware Approach for Detecting, Exposing and Tolerating High Level Atomicity Violations** DMTM'14
Orosa, Lois, Lourenco, Joao
(Workshop on Dependable Multicore and Transactional Memory Systems (DMTM'14))

Posters

- CODIC: A Low-Cost Substrate for Enabling Custom In-DRAM Functionalities and Optimizations** ISCA'22
Orosa, Lois, Wang, Yaohua, Sadrosadati, Mohammad, Kim, Jeremie, Patel, Minesh, Puddu, Ivan, Luo, Haocong, Razavi, Kaveh, Gomez-Luna, Juan, Hassan, Hasan, Mansouri, Nika Ghiasi, Ghose, Saugata, Mutlu, Onur
(9th IEEE International Symposium on Computer Architecture (ISCA'22),)
- Dataplant: A Low-Cost In-DRAM Value Generation Primitive for Enabling System Security Features** Retreat'18
Orosa, Lois, Wang, Yaohua, Puddu, Ivan, Sadrosadati, Mohammad, Hassan, Hasan, Tavakkol, Arash, Mansouri, Nika Ghiasi, Patel, Minesh, Kim, Jeremie, Gomez-Luna, Juan, Seshadri, Vivek, Azevedo, Rodolfo, Mutlu, Onur
(ETH Systems Group Industry Retreat)
- Improving GPU Power and Energy Efficiency** Retreat'18
Sadrosadati, Mohammad, Mirhosseini, Amirhossein, Ehsani, Seyed Borna, Ausavarungnirun, Rachata, Tavakkol, Arash, Falahati, Hajar, Orosa, Lois, Wang, Yaohua, Sarbazi-Azad, Hamid, Falsafi, Babak, Mutlu, Onur
(ETH Systems Group Industry Retreat)
- Processing-In-Memory Benchmark Suite and Analysis** Retreat'18
Gomez-Luna, Juan, Tavakkol, Arash, Boroumand, Amirali, Olivera, Geraldo F, Atamaner, Mert, Sadrosadati, Mohammad, Mansouri, Nika Ghiasi, Orosa, Lois, Mutlu, Onur
(ETH Systems Group Industry Retreat)
- A Case for an Amnesic DRAM Chip** Industry'17
Orosa, Lois, Wang, Yaohua, Sadrosadati, Mohammad, Hassan, Hasan, Tavakkol, Arash, Patel, Minesh, Seshadri, Vivek, Azevedo, Rodolfo, Mutlu, Onur
(ETH Industry day)
- AVPP: Address-first Value-next Predictor with Value Prefetching** Retreat'17
Orosa, Lois, Azevedo, Rodolfo, Mutlu, Onur
(ETH Systems Group Industry Retreat)
- FlexSig: Implementing Flexible Hardware Signatures** HiPEAC'12
Orosa, Lois, Antelo, Elisardo, Bruguera, Javier D
(7th HiPEAC Conference on High-Performance and Embedded Architectures and Compilers (HiPEAC'12),)

Honors and Awards

Top Pick in Hardware and Embedded Security 2025

2025

The paper Revisiting RowHammer: An Experimental Analysis of Modern DRAM Devices and Mitigation Techniques, was awarded a Top Pick in Hardware and Embedded Security 2025. The Top Picks award recognizes the best of the best in hardware security from among all hardware security papers published in the last six years, across all top security (e.g., IEEE S&P, USENIX SEC), architecture (e.g., ISCA, ASPLOS), computer-aided design (e.g., ICCAD, DAC, DATE) and hardware security (CHES) venues.

Top Picks

loisorosa.github.io/assets/pdf/honors_awards/TopPicks2025Certificate_Paper_2716.jpg

RYC2023-045785-I, Ramon y Cajal Tenure Track Contract

2023

Ranked 1st in Information and Communication Technologies area. Title More Capable, Secure and Efficient DRAM Main Memory Designs and Computer Architectures. Score 100/100, 251.800 EUR

Ministry of Science of Spain

[loisorosa.github.io/assets/pdf/](https://loisorosa.github.io/assets/pdf/honors_awards/RESOLUCION_ELEGIBLES_Y_%20RESERVAS_%20RYC_2023.pdf)

[honors_awards/RESOLUCION_ELEGIBLES_Y_%20RESERVAS_%20RYC_2023.pdf](https://loisorosa.github.io/assets/pdf/honors_awards/RESOLUCION_ELEGIBLES_Y_%20RESERVAS_%20RYC_2023.pdf)

R3 Certificate (established researcher)

2023

Certificate given to researchers who have developed a level of independence. Score 96/100.

Ministry of Science of Spain

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18 HiPEAC paper awards

Since 2017

I received 18 HiPEAC paper awards since 2017. The HiPEAC Paper Award aims to encourage HiPEAC members to publish their work at conferences in which Europe is not strongly represented. The award is given to a HiPEAC member who presents a full paper in one of the following list of conferences, ASPLOS, DAC, FCCM, HPCA, ISCA, MICRO, PLDI, POPL

HiPEAC

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Intel Hardware Security Academic Award 2022

2022

The paper BlockHammer: Preventing RowHammer at Low Cost by Blacklisting Rapidly-Accessed DRAM Rows was nominated as a finalist for the Intel Hardware Security Academic Award 2022 (one of 4 finalists out of 34 nominations)

Intel

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Services

Program Committees

ICS26, DSN24, DSN23, ASHES22, ASHES21

Skills

Programming Languages

C/C++, assembly, Python, Perl, Java, VHDL, Latex

Tools

Intel Instrumentation Pin Tool, DRAM Bender, SoftMC, Git

Simulators

Simics, GEMS, gem5, QEMU, Ramulator, Bochs, ZSim, DRAMSim, McPath, Spice, DRAMPower

Languages

Galician

Native speaker

Spanish

Native speaker

English

Fluent

Portuguese

Professional working proficiency